

What a Two-Byte Witness Buys on a Sprayed Fabric:

A Silicon-Measured Comparison of In-Fabric Grayhole Localization Against Passive Detection

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Abstract—Packet spraying spreads each flow of a machine-learning training job across every spine switch, so a link that silently drops a small fraction of its packets leaves its evidence divided among many paths. A detector that only counts arrivals then faces two limits: the deficit it looks for shrinks with the loss rate while the sampling noise of spraying does not, and an arrival deficit at a leaf is the product of an uplink and a downlink survival rate, so one vantage cannot tell the two directed links apart. The two most recent passive detectors for sprayed fabrics accept both limits. In this paper, we measure what an in-fabric per-directed-link witness buys against them. The witness is an existing primitive: two bytes per packet and two counters per directed link. We replay both passive detectors over one shared stream of sprayed traffic and measure the ledger on an Intel Tofino switch. The witness detects at a flat cost of about twenty-two million packets from a loss rate of 1.5 percent down to one in ten thousand, where the passive cost grows past one hundred million packets and then past the budget. It names the exact directed link where the passive detectors return the uplink and downlink pair, it recovers injected loss exactly on silicon with no false positive, and it costs about 0.14 percent of a 1400-byte payload. The advantage holds for several independent faults and fails under a fleet-wide shift in loss, where a relative passive test is better.

Index Terms—Data center networks, packet spraying, gray failure, network telemetry, programmable switches.

I. INTRODUCTION

Training fabrics for large machine-learning models spray packets. Instead of pinning a flow to one path, the leaf switch sends consecutive packets of the same flow to different spines, which balances load across a Clos fabric and is now the default of the Ultra Ethernet transport [1]–[3]. Spraying also changes what a failure looks like. A link that drops one packet in a thousand, or one in ten thousand, does not take a flow down. It slows every flow a little, because every flow crosses it. Such gray failures are common in production fabrics and hard to attribute [4], [5], and they can be costly for synchronous training jobs whose collective operations wait for the slowest packet [6]. We call a link that silently drops a small fraction of its packets a grayhole, and we call the problem of naming the directed link that carries it grayhole localization.

Grayhole localization on a sprayed fabric is challenging, due to two reasons. First, the evidence is divided. A flow that used to concentrate its loss on one path now spreads it across

k spines. Consequently, the per-path deficit a passive observer can see is $1/k$ of the fault, and it shrinks further as the loss rate falls, while the sampling noise of spraying, i.e., the variation in how many packets each spine happens to receive, does not shrink at all. Second, the evidence is aliased. A packet from one leaf to another crosses two directed links, an uplink into a spine and a downlink out of it. An arrival counter at the destination sees only the product of the two survival rates, and thus, a single vantage that counts arrivals can name a spine but cannot, by itself, tell the uplink from the downlink.

The two most recent detectors built for sprayed fabrics accept both limits. SprayCheck [7] applies a Z-test to per-spine arrival counts and resolves a directed link by intersecting reports from several leaves. FlowPulse [8] compares each ingress port's load against a learned baseline and applies a per-sender rule. Both are passive: they add nothing to the wire, they run on counters the switch already has, and they can be deployed today. However, a test on arrival counts needs enough packets for a deficit of fraction p to stand above spray noise, and that number grows at least as $1/p$. Both papers also state that one report can only flag the uplink and downlink pair [7], [8].

An in-fabric witness can remove both limits by construction. The sending leaf stamps every packet with a per-directed-link sequence number. The receiving switch keeps, per directed link, the highest sequence number seen and the number of packets seen. The difference of the two deltas is then the loss on that directed link and nothing else. This primitive is not new. NetSeer [9], LinkGuardian [10], LossRadar [11], and the link-layer retry of Ultra Ethernet [2] all place a per-link sequence witness in the data plane. What none of them reports is what the witness buys, and costs, on a sprayed fabric at the loss rates where the passive state of the art fails. That measurement, head-to-head against the passive detectors and on real silicon, is the contribution of this paper. We implement the witness as a receiver ledger on an Intel Tofino switch [12], with a two-byte stamp and two registers per directed link, and we treat it as the measured arm of a comparison rather than as a new design.

To make the comparison fair, we replay SprayCheck and FlowPulse from their own papers, with their own thresholds and their own localization rules. All three detectors read one shared stream of sprayed traffic, and each sees it only through the counters its own switch would have. We run fifty seeds per

loss rate from 1.5 percent down to 10^{-4} , and we report false positives next to every action rate. Where a passive detector ties or beats the witness, we report it as plainly as where it does not. Our contributions are as follows.

- **Measures the detection cost-scaling separation.** The witness detects a single-link grayhole on the first epoch after onset at every loss rate. Its median cost is a flat 22.0 million fleet packets from 1.5 percent to 10^{-4} , with an action rate of 1.00 and no false positive. SprayCheck's median cost grows from 24 million to 114 million packets as the loss rate falls to 10^{-3} , its action rate falls from 1.00 to 0.78, and at 10^{-4} it detects nothing within the same 160-million-packet budget. FlowPulse fails one order of magnitude earlier. We show that this is a scaling law of any arrival-count test and not an artifact of the budget (Section V).
- **Measures directed-link localization under spraying.** The witness names the exact directed link, and only it, at every loss rate and for both uplink and downlink faults. SprayCheck ties it at 1.0 to 1.5 percent loss, which we report as a null result, and degrades to the ambiguous uplink and downlink pair at 10^{-3} . FlowPulse cannot detect an uplink fault at any rate once several senders share the port (Section VI).
- **Measures the ledger on silicon.** On a Tofino switch the ledger recovers injected loss exactly from 10^{-2} down to 10^{-4} , i.e., six drops in sixty thousand packets, with zero false positives on more than forty thousand clean packets. The result holds for scheduled drops and for random per-packet drops, and a downlink grayhole is attributed to the downlink alone with zero false attribution on seven active sublinks (Section VII).
- **Bounds the claim with a correlated-fault gate.** Under two or three simultaneous independent faults the witness's advantage widens, because the passive intersection produces spurious links. Under a fleet-wide shift in loss with no single culprit the witness fails and flags all 64 directed links, and SprayCheck's relative test is the better tool. We report both, and we scope every other claim to independent per-directed-link faults (Section VIII).
- **Quantifies the cost.** The two-byte witness adds about 0.14 percent to a 1400-byte payload and six bytes of register state per directed link. It also costs one ingress pipeline stage, which brings the program to the twelve-stage ceiling of the Tofino 1 (Section IX).

To the best of our knowledge, this is the first head-to-head measurement of an in-fabric per-directed-link witness against the passive sprayed-fabric detectors, on the loss regime all three target, and on silicon. We claim no novelty for the witness itself. The per-link sequence stamp is the primitive of NetSeer and LinkGuardian, and the localization advantage we measure is a property of the information the witness holds rather than of a better inference. What the paper adds is the measurement, its cost, and its boundary.

The rest of the paper is organized as follows. Section II describes sprayed fabrics, grayholes, the fault model, and the objectives we evaluate. Section III describes the receiver ledger

as implemented. Section IV describes the replay harness, the two passive baselines, and the silicon testbed. Sections V to IX report detection cost, localization, the silicon measurements, the correlated-fault gate, and the cost. Section X discusses limitations, Section XI positions the work against prior systems, and Section XII concludes.

II. BACKGROUND, FAULT MODEL, AND OBJECTIVES

In this section, we first describe the sprayed leaf-spine fabric and the grayhole we study. Then we state the two limits of a passive vantage that the rest of the paper measures, and we define the fault model, the objectives, and the scope.

A. Sprayed Leaf-Spine Fabrics

A two-tier Clos fabric connects n leaf switches to k spine switches, and every leaf has one link to every spine. A packet from leaf L_a to leaf L_b crosses exactly two directed links: the uplink $L_a \rightarrow S_i$ and the downlink $S_i \rightarrow L_b$ for some spine S_i . We call each such one-way hop a *directed link*. A fabric with n leaves and k spines has $2nk$ of them. Under per-packet spraying the leaf chooses the spine for each packet independently, in the simplest case uniformly at random [1]. Consequently, a flow of N packets places about N/k packets on each of its k uplinks and on each of its k downlinks. The Ultra Ethernet specification adopts spraying as its default and adds a link-layer retry that can retransmit a lost frame across one link [2]. REPS extends spraying with a load balancer that steers away from paths that show loss [3]. Throughout the paper we use $n = 4$ and $k = 8$ in the replay harness, which is the topology SprayCheck evaluates [7], and a 4-leaf, 2-spine virtual fabric on silicon.

A *grayhole* is a directed link that forwards most of its packets and silently drops a fraction p of them. We study p from 1.5 percent down to 10^{-4} , and we call 10^{-3} to 10^{-4} the low-loss regime. A grayhole differs from a blackhole, which drops everything and is caught by any liveness check. It also differs from congestion loss, which is transient and is shared by every link behind the congested queue. Time is divided into *epochs*, i.e., the period at which a controller reads counters and decides. In the replay harness one epoch carries two million packets across the fleet, and the detectors act at epoch boundaries.

B. Two Limits of a Passive Vantage

A passive detector counts packets the switch already forwards and adds nothing to them. Under spraying it faces two limits, and both become worse as p falls.

The cost limit. A grayhole of rate p on one of k spines removes a fraction p of the packets that spine should have received. To notice the deficit, a detector compares that spine's count λ against the others. The comparison is only meaningful when the expected deficit λp exceeds the sampling noise of spraying, which is of order $\sqrt{\lambda}$ for independent spray decisions. Because a test with s standard deviations of margin needs $\lambda p > s\sqrt{\lambda}$, i.e., $\lambda > s^2/p^2$, the packets a count-based test needs grow at least as $1/p$, and for a Z-test as $1/p^2$. A

witness that observes each directed link's own transmit count does not compare against noise at all. It counts the missing packets directly, so its cost should not depend on p . Section V measures both.

The aliasing limit. An arrival counter at L_b for packets that came through spine S_i sees the product of the survival rates of $L_a \rightarrow S_i$ and $S_i \rightarrow L_b$. Consequently, one counter cannot tell which of the two directed links dropped the packets. SprayCheck resolves the pair by intersecting failure reports from several leaves that share the spine [7]. FlowPulse resolves it by asking whether every sender on the port is affected or only one [8]. Both rules need corroborating traffic, and that traffic can become scarce at low loss. A per-directed-link witness holds each link's own count and needs no corroboration. Section VI measures both.

C. Fault Model, Objectives, and Scope

We assume that the loss rate of a faulty directed link is stationary over the observation window and that faults on different directed links are independent. We argue that this is a reasonable assumption for the physical-layer causes that dominate gray failures in production, i.e., damaged optics, marginal cables, and misconfigured forward error correction. Each of these is local to one link and persists until repaired [5]. We assume that a healthy directed link has a small background loss rate, set to 10^{-5} in the harness, and we assume that one or several directed links can be faulty at once. We do not assume that the faulty link is known to carry any particular flow, and we do not assume that the controller can inject probes.

The paper evaluates four objectives, and we refer to them by label in the rest of the paper.

- **O1, detection cost.** The number of packets, and the fraction of trials, a detector needs to flag the faulty spine or link within a fixed budget, as a function of p , with the false-positive rate reported next to it.
- **O2, localization.** Whether the detector names exactly the faulty directed link, an ambiguous set that contains it, or nothing.
- **O3, silicon fidelity.** Whether the ledger, on a real switch, recovers the injected loss exactly at each p and attributes it to the injected directed link alone.
- **O4, the boundary.** Whether O1 and O2 survive several simultaneous faults, and whether they survive a fleet-wide, correlated shift in loss.

Scope. Two regimes are out of scope for the headline claims. Under a common-mode shift, in which every link's loss rises together, the ledger's decision rule fails, as Section VIII shows and Section X discusses. The claims of Sections V to VII are therefore stated for independent faults on a stationary fabric. Repairing or re-admitting a faulty link is not addressed; Section XI cites the systems that do. Blackholes are used only to validate the testbed and are not part of the evaluation.

III. THE RECEIVER LEDGER

The receiver ledger is the measured arm of this paper. It is our implementation of the per-link sequence witness that NetSeer [9] and LinkGuardian [10] place in the data plane, at

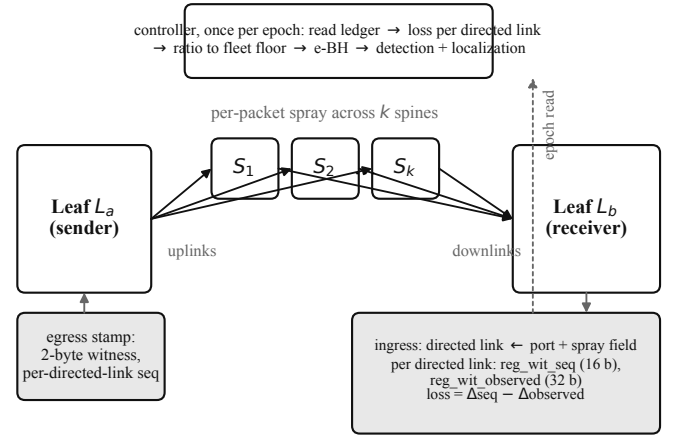


Fig. 1. The receiver ledger on a sprayed leaf-spine fabric. The sending leaf stamps each packet at egress with a two-byte per-directed-link sequence number. The receiving switch reconstructs the directed link from its own ingress port and the spray field, and keeps two registers per directed link: the last sequence number seen and the number of packets seen. Once per epoch the controller reads both registers, takes the difference of their deltas as the loss on that directed link, compares it with the fleet floor, and applies e-BH to name the faulty links.

reduced width, and we describe it here only far enough that the cost and the measurements of the later sections can be read. We claim no design contribution for it. In Figure 1, we show the data path from a sending leaf to a receiving leaf and the two places the ledger touches it.

Witness stamp. At the egress of the sending hop, the switch inserts a two-byte header carrying a 16-bit sequence number that it increments per directed sublink, i.e., per directed link and traffic class. The header carries nothing else. An earlier version of the ledger also carried a 16-bit link identifier, which doubled the wire cost. Because the receiving hop already knows its own ingress port, and because the spray field is already on the wire for load balancing, the identifier is redundant, and we removed it. Section IX reports what that removal cost in pipeline stages. The stamp is an in-fabric cost only: it is stripped before the packet reaches a host.

Link reconstruction. At the ingress of the receiving hop, a match table keyed on the ingress port and the spray field names the directed link, and a second table appends the traffic class, which the switch classifies fresh for every packet from the packet's size and its differentiated-services code point. Two tables are needed rather than one because the traffic class is a per-packet property, not a per-link one; a single table that baked a fixed class into each link entry would mislabel every packet whose class differed from the entry.

Ledger registers. For each of the $2nk$ directed links, and for each traffic class, the receiving switch keeps two registers: *seq*, the 16-bit sequence number of the last stamped packet seen on that sublink, and *obs*, a 32-bit lifetime count of the stamped packets seen. Both can be updated in the data plane at line rate, and both were on our switch. Over an epoch in which the two registers advance by Δseq , taken modulo 2^{16} , and Δobs , the loss on the sublink is

$$\ell = \Delta seq - \Delta obs, \quad (1)$$

because the sender advanced the sequence once per packet it sent and the receiver counted once per packet that arrived. Equation (1) is exact for in-order and for reordered arrivals. Because the sequence register keeps a frontier rather than a running sum, a late packet manufactures no phantom loss. A duplicated packet drives the estimate to -1 , which the controller can read as a duplicate. A packet lost after the last survivor of an epoch is counted in the next epoch, when the next survivor advances the frontier. We verified all three properties against the compiled program on the vendor's software model before any hardware measurement (Section IV).

Decision. Once per epoch the controller reads every sublink's registers, forms the loss ratio ℓ/Δ_{seq} , and compares it with the fleet floor, a leave-one-out estimate of the background loss rate that excludes the sublink under test. Because a healthy sublink's ratio sits at the floor while a faulty one sits far above it, the controller can test each sublink on its own evidence. It never compares a sublink against the sampling noise of the other spines. The per-sublink evidence is accumulated as an e-process. The set of sublinks whose evidence exceeds the e-BH threshold at a chosen false discovery rate [13] is the set the controller names as faulty. That set is both the detection and the localization, because a rejected sublink is a directed link and only the link that lost packets can have an anomalous ratio. We note that this decision rule is the part of the ledger that fails under a fleet-wide shift in loss, because the floor then moves with the fault (Section VIII).

IV. METHODOLOGY

The evaluation has two vehicles. A replay harness compares the ledger with the two passive detectors on one shared stream of sprayed traffic, because the passive detectors are host-side systems that cannot run on our fabric and because a head-to-head needs every arm to see the same packets. A silicon testbed then measures the ledger's own side of that comparison on a real switch. In this section, we describe the harness, the three arms, the metrics, and the testbed.

A. The Replay Harness

Shared stream. Every epoch, the harness draws one per-packet spray decision and one per-hop survival outcome for every packet, and it feeds the same draw to all three arms. Spray is independent and uniform across the $k = 8$ spines, which matches SprayCheck's own model and FlowPulse's temporal-symmetry assumption [7], [8]. Survival is independent per directed link with rate $1 - p$ on a faulty link and $1 - 10^{-5}$ on a healthy one. Each arm sees only what its own switch would see. The ledger arm receives the exact per-sublink transmit and receive counts its registers produce. The SprayCheck arm receives receive-only per-spine arrival counts; no transmit count and no drop count exists in its interface, so none can be passed to it by mistake. The FlowPulse arm receives receive-only per-port load, predicted from its own bootstrapped history, and never the ledger's transmit ground truth.

Scenario. Each trial runs ten epochs of clean traffic, during which FlowPulse bootstraps its baseline and the ledger warms

its floor. The trial then degrades one spine, or one directed link, to the target rate p for the rest of the run. For detection (O1) the fleet carries two million packets per epoch, and a trial ends when the arm flags the faulty spine or after 80 post-onset epochs, a budget of 160 million packets. For localization (O2) the harness extends to the two-hop fabric of $n = 4$ leaves, and each ordered leaf pair carries two million packets per epoch. The localization budget is 60 post-onset epochs, which is deliberately generous to the passive arms so that SprayCheck can complete a corroborating round of measured flows. For the correlated gate (O4) the window is 40 post-onset epochs. We run 50 seeds per cell.

Two fidelity checks that changed the harness. Before trusting any number we checked each arm against its own paper. Two checks failed and were fixed. First, a per-packet traffic generator made one two-million-packet epoch take tens of seconds; we vectorized it with a multinomial draw, which changed no distribution. Second, at an initial volume of 200 thousand packets per epoch, FlowPulse's fixed one-percent threshold false-alarmed on a healthy spine in every trial. The reason is that the relative spray noise at that volume, about 0.6 percent, sat close to its threshold. We raised the volume to two million packets per epoch, at which its false-positive rate on healthy spines is 0.0000, consistent with the paper's own claim of under one percent [8]. We report that rate next to every result.

B. The Three Arms

The ledger. The controller of Section III, unmodified from the version measured on silicon, runs over the exact per-sublink counts. Its localized set is the set of sublinks e-BH rejects.

SprayCheck-Z. We reimplement SprayCheck's detector from its paper [7]: a Z-test on per-spine arrival counts with sensitivity s calibrated under the paper's own Poisson model at 2.5 million packets per spine, which gives $s = 3.24$. For localization we implement its full cross-leaf rule: a report flags the uplink and downlink pair of a path, and a single link is named only when it lies in the intersection of reports from different leaves. SprayCheck's real spraying uses join-the-shortest-of-two-queues, which has less noise than the Poisson model; a sensitivity tuned to that would shift its cost curve down by a constant factor but not change how the cost scales with p .

FlowPulse- θ . We reimplement FlowPulse's port-load detector with its fixed one-percent threshold against a learned per-port baseline, and its per-sender localization rule: if every sender on an ingress port is affected the local downlink is named, and if only one sender is affected the uplink from that sender's leaf is named [8]. We give the arm per-sender visibility on every port, which its paper assumes; this is the charitable reading, not a weakened one.

C. Metrics

Action rate is the fraction of the 50 seeds in which the arm flags the faulty spine or link within the budget, with a 95

percent Wilson interval [14]. **Packets to detect** is the fleet-wide packet count from the fault's onset to the first correct flag, reported as a median with its interquartile range over the seeds that detected; we mark it as survivorship-biased wherever the action rate is below one. **False positive** is any healthy spine or link flagged, reported next to every action rate. For O2, **exact** means the arm named the true directed link and only it, **ambiguous** means it named a set that contains a healthy link, and **miss** means it named nothing. We also report the mean set size over detections and a paired exact McNemar test between arms on the same seeds. For O4, **recall** is the fraction of true faulty links named, and **exact-all** is the fraction of seeds that named the true set and nothing else. We report the false-positive rate both over the whole window and at its final epoch, so that a transient false flag can be separated from a persistent one.

D. The Silicon Testbed

Switch. The ledger runs on an Intel Tofino 1 switch [12] with SDE 9.13.2. A 4-leaf, 2-spine virtual fabric is emulated on the switch itself: a four-lane 25 Gb/s direct-attach cable loops four front-panel ports back to the switch, and each directed virtual link is one traffic-manager queue on one loop port. With four traffic classes, the program tracks 1024 sublinks. The pipeline occupies 12 ingress and 5 egress match-action stages (Section IX).

Traffic and ground truth. A server with a 25 Gb/s network interface sends 1400-byte packets at 20 thousand packets per second, and a differentiated-services code point selects the traffic class and thus the sublink. Two fault injectors run in the switch's egress pipeline. The exact injector drops exactly D stamped packets on a chosen sublink, so the ground truth is D . The stochastic injector draws a 16-bit random value per packet and drops the packet when the value falls below $W = \lfloor p \cdot 2^{16} \rfloor$, so it drops each packet with probability $W/2^{16}$. For this injector the ground truth is its own direct counter of packets dropped, read from the switch, because a random injector has no preset count. Every cell reads a baseline census immediately before arming and reports deltas against it. Every cell also keeps Δ_{seq} below 2^{16} , so that the sequence wraps at most once, and clears both injectors before and after. The recovered loss is Equation (1) read through the controller's census.

Model validation. Before any hardware run, the compiled program passed nine correctness tests on the vendor's software model. They cover the exact recovery of injected loss, the invisibility of a trailing loss until the next survivor, the 32-bit transmit frontier past 255 and 256 packets, the absence of phantom loss under reordering, the -1 signature of a duplicate, and the realized rate of the stochastic injector, which fell within four standard deviations of its target. The hardware measurements of Section VII were then taken on the same build.

V. DETECTION COST (O1)

We expect the ledger's cost to be flat in p and the passive detectors' cost to grow as $1/p$ (Section II-B). In Figure 2 and

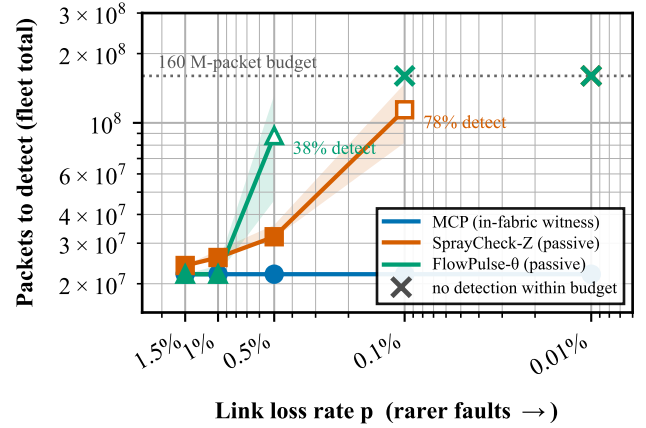


Fig. 2. Median packets to detect a single-spine grayhole against loss rate p , over 50 seeds, $k = 8$ spines, and a budget of 160 million packets (dotted ceiling). Filled markers are cells with an action rate of 1.00; hollow markers are annotated with the fraction of seeds that detected, and their median is over those seeds only; a cross on the ceiling marks a cell in which no seed detected. The ledger's cost is flat, the passive detectors' cost grows toward the ceiling and then crosses it.

TABLE I
DETECTION OF A SINGLE-SPINE GRAYHOLE WITHIN A 160-MILLION-PACKET BUDGET, 50 SEEDS PER CELL: ACTION RATE (ACT.) AND, OVER THE SEEDS THAT DETECTED, THE MEDIAN FLEET PACKETS TO THE FIRST CORRECT FLAG (PKTS). THE 95 PERCENT WILSON INTERVAL IS [0.93, 1.00] FOR EVERY 1.00, [0.65, 0.87] FOR 0.78, [0.26, 0.52] FOR 0.38, AND [0.00, 0.07] FOR EVERY 0.00. FALSE POSITIVES WERE 0.00 FOR EVERY ARM AT EVERY RATE.

p	Ledger		SprayCheck-Z		FlowPulse-0	
	Act.	Pkts	Act.	Pkts	Act.	Pkts
1.5%	1.00	22.0 M	1.00	24.0 M	1.00	22.0 M
1.0%	1.00	22.0 M	1.00	26.0 M	1.00	22.0 M
0.5%	1.00	22.0 M	1.00	32.0 M	0.38	88.0 M
10^{-3}	1.00	22.0 M	0.78	114.0 M	0.00	—
10^{-4}	1.00	22.0 M	0.00	—	0.00	—

Table I, we show the packets each arm needed to flag the faulty spine and the fraction of seeds in which it did so, for five loss rates from 1.5 percent to 10^{-4} ; Figure 3 shows the action rates with their intervals.

The ledger's cost is flat. The ledger flagged the faulty spine in every seed at every rate, with a median of 22.0 million fleet packets and an interquartile range of 22 to 24 million, and it did so on the first epoch after onset in every case. The reason is that the quantity it tests is a loss ratio relative to the fleet floor, and a fault at any p that is far above the floor of 10^{-5} clears the test on one epoch's worth of that link's traffic. Its cost is set by the epoch volume, not by p . Its false-positive rate was 0.00.

The passive cost grows and the action rate collapses. SprayCheck-Z matched the ledger at 1.5 percent and needed 24, 26, and 32 million packets at 1.5, 1.0, and 0.5 percent. At 10^{-3} its median rose to 114 million packets, five times the ledger's, and that median is over the 78 percent of seeds that detected at all; the interquartile range widened from 24 to 24 million at 1.5 percent to 82 to 148 million at 10^{-3} . At

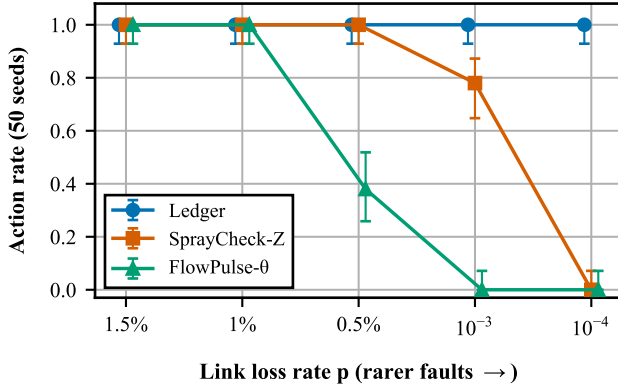


Fig. 3. Action rate against loss rate with 95 percent Wilson intervals, 50 seeds per cell, within the 160-million-packet budget; the false-positive rate was 0.00 for every arm at every rate.

10^{-4} no seed detected within the budget. FlowPulse- θ failed one order of magnitude earlier: it matched the ledger at 1.5 and 1.0 percent, detected in 38 percent of seeds at 0.5 percent with a median of 88 million packets, and detected nothing at 10^{-3} and below. Both passive arms also had a false-positive rate of 0.00, so the collapse is a loss of sensitivity, not a trade against false alarms.

This is a scaling law, not a budget artifact. Because a Z-test on a spine's arrival count λ separates a deficit of fraction p from spray noise only when $\lambda > s^2/p^2$ (Section II-B), its packets to detect grow at least as $1/p$. The separation is visible inside the budget: at 10^{-3} SprayCheck already needed five times the ledger's packets for the seeds it caught, and the 22 percent it missed at 10^{-3} and the whole column at 10^{-4} are the same wall pushed past 160 million packets. A larger budget would move the passive action-rate floor to a rarer p , but it cannot flatten the curve, and the ledger's flat line should not depend on the budget at all, because it detects on the first epoch. We note that this is a statement about cost, not capability: a passive detector with an unbounded budget can eventually detect at 10^{-4} , and nothing here says otherwise. Within any fixed operational budget, however, the passive action rate collapses as faults become rarer, and the witness's does not.

Two disclosures. SprayCheck's sensitivity is calibrated under its paper's Poisson model; its real join-the-shortest-queue spraying has less noise, and a sensitivity tuned to it would lower its curve by a constant factor without changing the slope. The conditional medians of the hollow points in Figure 2 are survivorship-biased, which is why the figure annotates each with its action rate and why we read the action-rate collapse, not the slope of a censored median, as the load-bearing signal.

VI. LOCALIZATION (O2)

Detection answers whether something is wrong; localization answers which directed link, which is the question a repair needs. We expect the ledger to name exactly one directed link at every rate, because it holds each link's own count, and we

TABLE II
LOCALIZATION OF A SINGLE DIRECTED-LINK GRAYHOLE ON A 4-LEAF, 8-SPINE FABRIC, 50 SEEDS PER CELL, GENEROUS BUDGET OF 60 POST-ONSET EPOCHS. EXACT: NAMED THE TRUE DIRECTED LINK AND ONLY IT (95 PERCENT WILSON INTERVAL). AMB.: NAMED A SET CONTAINING A HEALTHY LINK. MISS: NAMED NOTHING. SIZE: MEAN SET SIZE OVER DETECTIONS. THE LEDGER SCORED EXACT 1.00 [0.93, 1.00] WITH SIZE 1.00 IN EVERY CELL OF BOTH FAMILIES AND IS OMITTED FROM THE ROWS.

p	Arm	Exact	Amb.	Miss	Size
<i>Downlink fault ($S_0 \rightarrow L_0$)</i>					
1.5%	SprayCheck-Z	1.00 [0.93, 1.00]	0.00	0.00	1.00
	FlowPulse- θ	0.94 [0.84, 0.98]	0.06	0.00	1.18
1.0%	SprayCheck-Z	0.96 [0.87, 0.99]	0.04	0.00	1.04
	FlowPulse- θ	0.22 [0.13, 0.35]	0.78	0.00	2.74
0.5%	SprayCheck-Z	0.34 [0.22, 0.48]	0.66	0.00	1.70
	FlowPulse- θ	0.00 [0.00, 0.07]	0.00	1.00	—
10^{-3}	SprayCheck-Z	0.00 [0.00, 0.07]	1.00	0.00	2.00
	FlowPulse- θ	0.00 [0.00, 0.07]	0.00	1.00	—
10^{-4}	SprayCheck-Z	0.00 [0.00, 0.07]	0.26	0.74	2.00
	FlowPulse- θ	0.00 [0.00, 0.07]	0.00	1.00	—
<i>Uplink fault ($L_1 \rightarrow S_0$)</i>					
1.5%	SprayCheck-Z	1.00 [0.93, 1.00]	0.00	0.00	1.00
	FlowPulse- θ	0.00 [0.00, 0.07]	0.00	1.00	—
1.0%	SprayCheck-Z	1.00 [0.93, 1.00]	0.00	0.00	1.00
	FlowPulse- θ	0.00 [0.00, 0.07]	0.00	1.00	—
0.5%	SprayCheck-Z	0.26 [0.16, 0.40]	0.74	0.00	1.78
	FlowPulse- θ	0.00 [0.00, 0.07]	0.00	1.00	—
10^{-3}	SprayCheck-Z	0.02 [0.00, 0.10]	0.92	0.06	1.98
	FlowPulse- θ	0.00 [0.00, 0.07]	0.00	1.00	—
10^{-4}	SprayCheck-Z	0.00 [0.00, 0.07]	0.26	0.74	2.00
	FlowPulse- θ	0.00 [0.00, 0.07]	0.00	1.00	—

expect the passive arms to degrade to the uplink and downlink pair once corroborating traffic runs short (Section II-B). Table II presents the outcome for a downlink fault and for an uplink fault, over 50 seeds per cell on the two-hop fabric of Section IV, and Figure 4 plots the exact-localization rates and the size of the named set.

The ledger names exactly one directed link. In every cell of both families the ledger's exact rate was 1.00 [0.93, 1.00], its mean set size was exactly 1.00, and it named a healthy link in no seed. This follows from two facts already on record: it detects in every seed with a false-positive rate of 0.00 (Section V), and the sublink it rejects is a directed link, so only the link that lost packets has an anomalous ratio. We state plainly that this score is close to definitional. The ledger is the only arm that sees each directed link's own transmit count, and the arms do not have equal information by design. The measured result is that the witness removes the aliasing the passive arms cannot, not that a better inference beat them on equal evidence.

SprayCheck-Z ties at high loss and degrades to the pair at low loss. At 1.5 percent in both families and at 1.0 percent for the uplink, SprayCheck's cross-leaf intersection resolved the fault to one directed link as well as the ledger did; the paired difference was 0.00 with $p = 1.0$ on the McNemar test, and we report it as a null result. At 0.5 percent its exact rate fell to 0.34 for the downlink and 0.26 for the uplink. At 10^{-3} it fell to 0.00 and 0.02 with a mean set size of 2.00, i.e.,

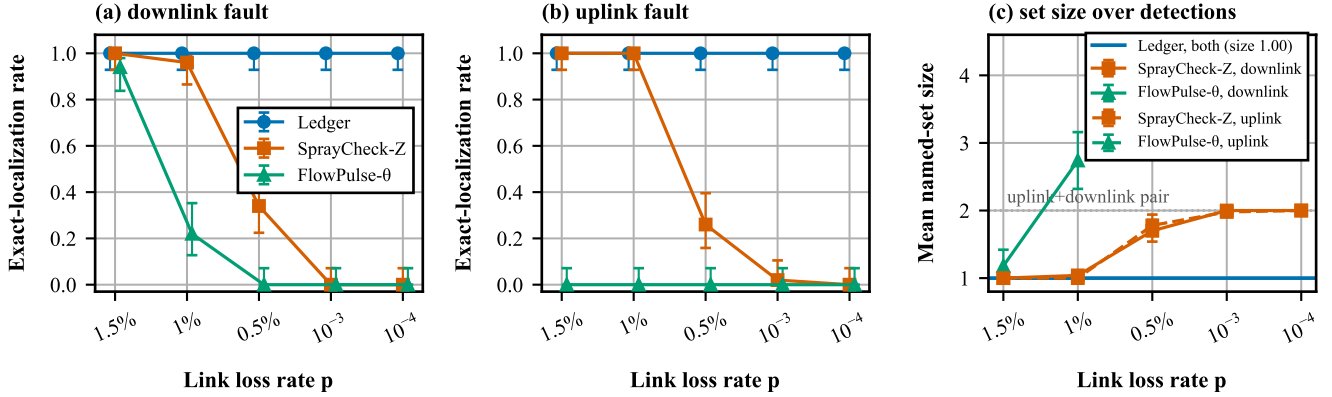


Fig. 4. Localization under spraying, 50 seeds per cell with 95 percent intervals. (a) Exact-localization rate for a downlink fault. (b) Exact-localization rate for an uplink fault. (c) Mean size of the named set over detections: 1 is the exact directed link, 2 is the uplink and downlink pair; the ledger's set size is 1.00 in every cell, and a missing point means the arm detected nothing.

the uplink and downlink pair its paper says an uncorroborated report is left with. The reason is that at those rates its Z-test flags the faulty spine on too few distinct leaves to complete the intersection, so it can name the spine but not the directed link. Every such pair contains one healthy link, which is why the ambiguous column reaches 1.00 there: a repair that acted on the set would touch the wrong hop half the time. At 10^{-4} it missed in 74 percent of seeds even under the generous budget.

FlowPulse- θ cannot localize an uplink fault at any rate. For a downlink fault it was exact in 94 percent of seeds at 1.5 percent, fell to 22 percent at 1.0 percent with a mean set size of 2.74, because its all-or-one rule landed in neither branch and returned the port's whole candidate set, and missed at 0.5 percent and below. For an uplink fault it missed in every seed at every rate, including 1.5 percent. The reason is that a single sender's loss moves the aggregate load of the ingress port by only about $1/(\text{number of senders})$ of the drop rate, here about 0.5 percent for a 1.5 percent uplink loss shared by three senders, which never clears the paper's fixed one-percent threshold. This is faithful to FlowPulse's own design: in its native setting one sender occupies each port and an uplink fault is the whole port's load. Under multi-sender spraying the same fault is diluted below its threshold. The gap is a property of the sprayed fabric, reproduced, not a handicap imposed on the baseline.

Boundary. These results hold for one independent fault on a fabric of four leaves and eight spines with uniform spray. FlowPulse's dilution grows with the number of senders per port, so it is worse at larger radix; SprayCheck's intersection improves with more leaves but needs a longer corroborating round. We expect the ordering of the arms to survive both changes, while the exact crossover rates are properties of this fabric, not universal constants.

VII. THE LEDGER ON SILICON (O3)

The replay harness models the ledger's counters. This section measures them. Because the passive detectors are host-side systems that cannot run on our fabric, silicon cannot host the head-to-head; what it can do is show that the ledger's own

TABLE III
RECOVERY OF INJECTED LOSS ON THE TOFINO SWITCH, ONE SUBLINK, EXACT INJECTOR: PACKETS SENT N , DROPS ARMED D , REGISTER DELTAS, AND THE RECOVERED LOSS $\Delta_{seq} - \Delta_{obs}$. EVERY CELL RECOVERED D EXACTLY, AND THE TWO CLEAN CELLS RECOVERED 0.

D/N	N	D	Δ_{seq}	Δ_{obs}	Recovered	Match
clean	20 000	0	20 000	20 000	0	yes
clean	20 000	0	22 050	22 050	0	yes
10^{-2}	5 000	50	5 000	4 950	50	yes
10^{-2}	5 000	50	5 000	4 950	50	yes
10^{-3}	30 000	30	30 000	29 970	30	yes
10^{-3}	30 000	30	30 000	29 970	30	yes
5×10^{-4}	40 000	20	40 000	39 980	20	yes
10^{-4}	60 000	6	60 000	59 994	6	yes
10^{-4}	60 000	6	60 000	59 994	6	yes

side of the comparison is measured rather than modelled, at the same loss rates, on a real switch. We expect the ledger to recover the injected loss exactly at every rate, on the first read after the traffic, and to attribute it to the injected directed link alone; a one-packet discrepancy may appear where a loss trails the last survivor of a read.

Recovery across the loss regime. Table III and Figure 5(a) present nine cells on one uplink sublink. In every cell the recovered loss equals the armed drop count exactly, from 50 drops in 5 000 packets down to 6 drops in 60 000, and the two clean cells, together more than 42 000 packets, recovered 0. The ledger reported each value on the first census read after the traffic. It does not accumulate packets toward a statistical threshold, because Equation (1) counts the missing packets rather than inferring them. This is the structural contrast with Section V. At 10^{-4} , where the passive detectors detect nothing within the budget, the witness recovers the six lost packets on real hardware.

Random drops. A scheduled burst is the easy case, so we repeated the measurement with the stochastic injector, which drops each packet independently with probability $W/2^{16}$, and we compared the recovered loss with the injector's own counter of packets it dropped. At a target of 10^{-4} ($W = 7$, realized rate 1.07×10^{-4}) on 60 000 packets, the ledger

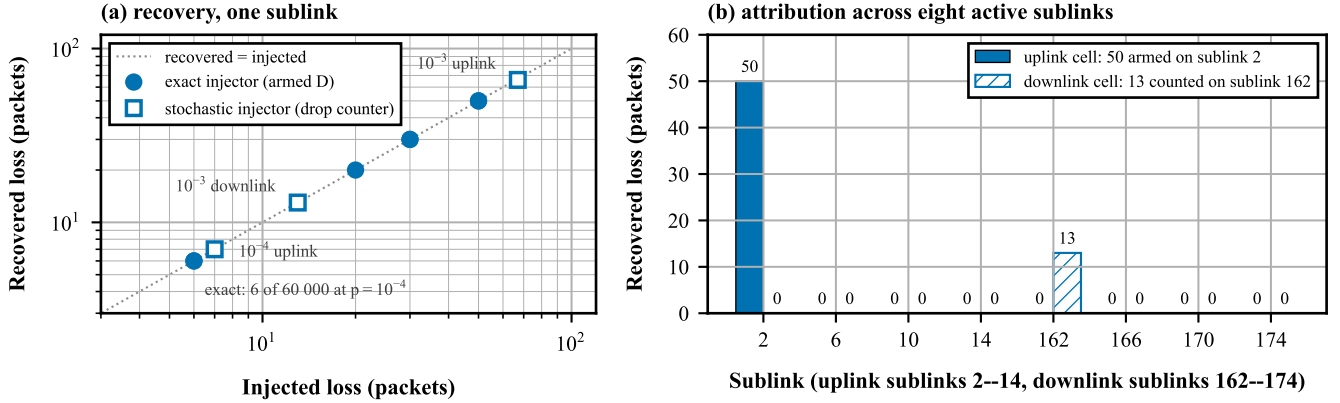


Fig. 5. The ledger on the Tofino switch. (a) Recovered loss against injected loss on one sublink, on logarithmic axes, for the exact injector, whose ground truth is the armed count, and for the stochastic injector, whose ground truth is its own drop counter; every point lies on the identity except the 10^{-3} uplink cell, one packet below it. (b) Recovered loss on each of the eight active sublinks for the uplink cell and for the downlink cell; only the injected sublink recovers a non-zero loss in each.

recovered 7 and the injector counted 7. At a target of 10^{-3} ($W = 66$, realized 1.01×10^{-3}) the ledger recovered 66 and the injector counted 67. Two facts in the data account for the one-packet gap, and we believe neither is a recovery error. First, a drop with no later survivor on the sublink is not exposed until the next survivor advances the frontier, which is the trailing-loss property of Section III. Second, on that uplink egress the injector's offered count ran about 180 to 190 packets above the 60 000 host departures. In other words, a small population of packets that are not part of the host-stamped sequence crossed the same egress, and the injector saw them but the ledger did not. The recovered value tracks the loss of stamped packets, which is the quantity the ledger is defined to recover.

Localization on the uplink. Figure 5(b) shows both attribution cells. We armed 50 drops on one uplink sublink and sent 20 000 packets across all four traffic classes, 5 000 each, so that every one of the eight tracked sublinks carried traffic. The injected sublink recovered 50, and the other seven recovered 0. Among the seven is the downlink sublink of the same traffic class, which showed 4 950 arrivals rather than 5 000, because the 50 packets dropped on the uplink never reached the downlink to be stamped. It manufactured no loss there, since its Δ_{seq} and Δ_{obs} advanced together.

Localization on the downlink. The mirror case is the one the passive vantage cannot resolve. We armed the stochastic injector at 10^{-3} on a downlink sublink and sent 80 000 packets across all four classes, 20 000 each. The downlink sublink recovered 13 and the injector counted 13. Every other sublink recovered 0, including the uplink sublink of the same class, whose 20 000 packets had all been witnessed before the downlink drop. On silicon, then, a downlink grayhole is attributed to the downlink directed link alone and an uplink grayhole to the uplink alone, with zero false attribution on seven active sublinks in each case. This is the disaliasing property the replay harness measured in Section VI.

Scope. These cells measure recovery fidelity and false attribution for one independent fault at a time on a 4-leaf,

2-spine virtual fabric. The recovered loss is magnitude-based and is the same for a burst and for dispersed drops of the same count; the one-gap-per-discontinuity property is covered by the model tests of Section IV, not here. The added-load figure of Section IX is computed from the header size, not measured as saturating throughput.

VIII. THE BOUNDARY: CORRELATED FAULTS (O4)

Every result so far injects one independent fault. This section asks whether the advantage survives several faults at once and whether it survives a correlated, fleet-wide shift in loss. It is a gate rather than a pass: if the advantage collapses under correlation, that bounds the headline claims to the independent regime, which is the pre-registered scope, and we would rather know. The harness of Section IV drives three regimes over the 64 directed links of the two-hop fabric, 50 seeds per cell and 40 post-onset epochs, with the ledger's controller unmodified. Two anchors accompany every cell: a do-nothing arm that never acts, and an oracle that returns the true set. The injected loss was verified in the data of every cell, not in the configuration. Figure 6 summarizes the three regimes; Tables IV and V give the full numbers.

A. Several Independent Faults

Table IV presents the first regime, in which $M \in \{2, 3\}$ distinct directed links are faulty at the same rate against a healthy background. The ledger named every one of the faulty links on the first epoch after onset, with no false link, at both 0.5 percent and 10^{-3} . The reason is that it judges each directed link on that link's own evidence, so a second fault elsewhere in the fabric changes nothing about the first. SprayCheck-Z became worse as faults multiplied. Because several simultaneous failure reports create spurious cross-leaf intersections, it recalled the true faults at 0.5 percent but added 1.2 to 1.4 healthy links per seed. Consequently, it named exactly the true set in only 18 to 28 percent of seeds. At 10^{-3} its recall itself fell to 0.63 to 0.75, and a healthy link was still

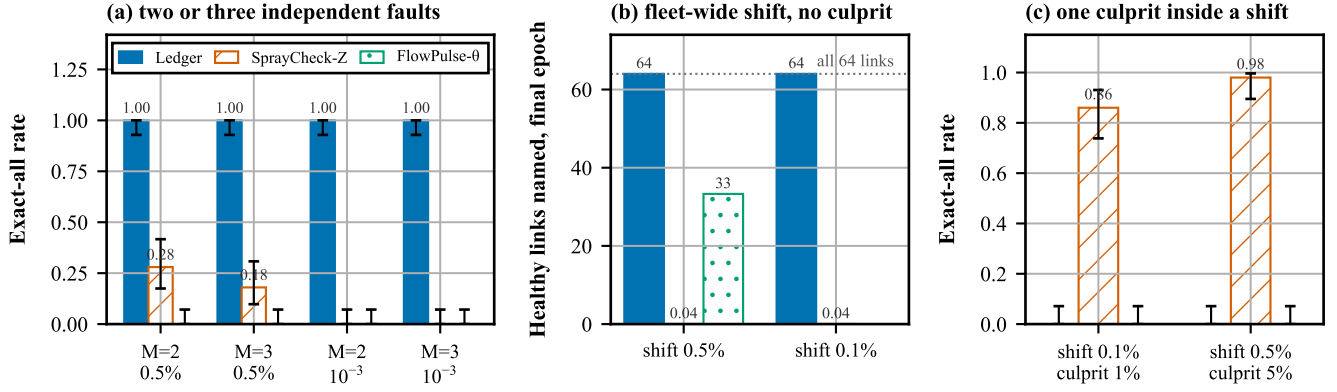


Fig. 6. The correlated-fault gate, 50 seeds per cell over 40 post-onset epochs on 64 directed links. (a) Exact-all rate, with 95 percent intervals, for two and three simultaneous independent faults. (b) Healthy links falsely named at the final epoch under a fleet-wide shift with no culprit, where the correct answer is zero and 64 is every link. (c) Exact-all rate for one culprit embedded in a shift. The ledger holds in (a) and fails in (b) and (c), where SprayCheck-Z holds.

TABLE IV

TWO OR THREE SIMULTANEOUS INDEPENDENT FAULTS (R1), 50 SEEDS PER CELL. RECALL: FRACTION OF TRUE FAULTY LINKS NAMED. EXACT-ALL: FRACTION OF SEEDS THAT NAMED THE TRUE SET AND NOTHING ELSE. FP: FRACTION OF SEEDS THAT NAMED A HEALTHY LINK, OVER THE WHOLE 40-EPOCH WINDOW / AT THE FINAL EPOCH. FALSE: MEAN HEALTHY LINKS NAMED OVER THE WINDOW. FLOWPULSE- θ NAMED NOTHING IN THESE CELLS (RECALL 0.00) AND IS OMITTED.

Cell	Arm	Recall	Exact-all	FP (win./fin.)	False
$M=2, 0.5\%$	Ledger	1.00	1.00	0.00 / 0.00	0.00
	SprayCheck-Z	1.00	0.28	0.72 / 0.00	1.20
$M=3, 0.5\%$	Ledger	1.00	1.00	0.00 / 0.00	0.00
	SprayCheck-Z	1.00	0.18	0.82 / 0.02	1.44
$M=2, 10^{-3}$	Ledger	1.00	1.00	0.00 / 0.00	0.00
	SprayCheck-Z	0.75	0.00	0.98 / 0.52	2.54
$M=3, 10^{-3}$	Ledger	1.00	1.00	0.00 / 0.00	0.00
	SprayCheck-Z	0.63	0.00	0.96 / 0.54	2.58

named at the final epoch in about half the seeds. FlowPulse- θ missed at these rates, as in the single-fault sweep. The independent-fault advantage is therefore stronger with several faults than with one, and the spurious-intersection behaviour is a limitation of the passive localizer that we have not found reported elsewhere.

B. A Fleet-Wide Shift With No Culprit

The upper part of Table V presents the second regime, in which every directed link's loss rises to the same level and no link is individually at fault, so the correct answer is to name nothing. The ledger failed this gate. It named all 64 directed links in every seed, on every epoch through the end of the window, which is worse than doing nothing. The reason is that its fleet floor is a leave-one-out estimate that lags a shift shared by every link. Each link's ratio then reads the shifted but healthy rate as strong evidence, and e-BH rejects nearly every link. The rejected links leave the pool from which the floor is estimated, and the fleet latches. We had predicted this failure from our own analysis of the decision rule before the gate was run, and the gate reproduced it on fresh seeds. SprayCheck-Z was the arm that held here, with a false-positive rate of 0.02 at the final epoch. Its Z-test normalizes each flow by that flow's

TABLE V

A FLEET-WIDE SHIFT IN LOSS WITH NO SINGLE CULPRIT (R2, GROUND TRUTH IS THE EMPTY SET) AND ONE CULPRIT EMBEDDED IN THE SHIFT (R3). FOR R2 THE COLUMNS ARE THE FALSE-POSITIVE RATE OVER THE WINDOW / AT THE FINAL EPOCH AND THE MEAN NUMBER OF HEALTHY LINKS NAMED OVER THE WINDOW / AT THE FINAL EPOCH. FOR R3 THEY ARE RECALL OF THE CULPRIT, EXACT-ALL, FALSE-POSITIVE RATE OVER THE WINDOW / AT THE FINAL EPOCH, AND MEAN HEALTHY LINKS NAMED OVER THE WINDOW. 50 SEEDS PER CELL; DO-NOTHING SCORES 0.00 ON EVERY R2 COLUMN.

R2 cell	Arm	FP (win./fin.)	False (win./fin.)
shift 0.5%	Ledger	1.00 / 1.00	64.0 / 64.0
	SprayCheck-Z	0.20 / 0.02	0.48 / 0.04
	FlowPulse- θ	1.00 / 1.00	64.0 / 33.3
shift 0.1%	Ledger	1.00 / 1.00	64.0 / 64.0
	SprayCheck-Z	0.28 / 0.02	0.68 / 0.04
	FlowPulse- θ	0.00 / 0.00	0.00 / 0.00

R3 shift / culprit	Arm	Recall	Exact-all	FP (win./fin.)	False
0.1% / 1%	Ledger	1.00	0.00	1.00 / 1.00	63.0
	SprayCheck-Z	1.00	0.86	0.14 / 0.00	0.22
	FlowPulse- θ	1.00	0.00	1.00 / 0.50	3.0
0.5% / 5%	Ledger	1.00	0.00	1.00 / 1.00	63.0
	SprayCheck-Z	1.00	0.98	0.02 / 0.00	0.02
	FlowPulse- θ	1.00	0.00	1.00 / 1.00	33.9

own observed count, which falls with the shift, so a uniform shift can cancel. FlowPulse- θ flagged nearly every port when the shift's two-hop deficit crossed its one-percent threshold and flagged nothing when it did not; below the threshold it is blind rather than resistant.

C. A Culprit Inside the Shift

The lower part of Table V presents the third regime, a shift plus one link worse than the shifted background. The ledger did find the culprit, in every seed and on the first epoch, but it drowned the culprit among the other 63 links, so its exact-all rate was 0.00 and its answer was not actionable. SprayCheck-Z isolated the culprit cleanly against the shifted background, with exact-all rates of 0.86 and 0.98 and no healthy link named at the final epoch, for the same reason that let it hold in the second regime. In this regime, then, the exact directed-link

TABLE VI

COST OF THE LEDGER BESIDE THE SELF-REPORTED COST OF THE PASSIVE DETECTORS. LEDGER NUMBERS ARE FROM THE HARDWARE COMPILE ON SDE 9.13.2, WHICH MATCHED THE 9.13.1 COMPILE EXACTLY; THE 4-BYTE COLUMN IS THE EARLIER VERSION THAT ALSO CARRIED A LINK IDENTIFIER. SPRAYCHECK'S STATE IS UNDER 2 KB IN TOTAL BUT COVERS ONE FLOW AT A TIME, ROTATING ABOUT ONCE A MINUTE; FLOWPULSE'S PAPER STATES NO MEMORY FIGURE. WE HAVE NOT REIMPLEMENTED EITHER IN P4, SO NO STAGE COUNT EXISTS FOR THEM.

	Ledger, 2 B	Ledger, 4 B	Passive
Wire bytes per packet	2	4	0
Added load (1400 B)	0.14%	0.28%	0
State per directed sublink	6 B	6 B	see caption
Sublinks covered at once	1024	1024	one flow / port
Ingress / egress stages	12 / 5	11 / 5	—
SRAM / TCAM blocks	91 / 16	89 / 15	—

localization that is the ledger's advantage under independent faults becomes SprayCheck's advantage.

D. What the Gate Decides

The advantage holds, and widens, for several simultaneous independent faults, and it fails under a common-mode shift. Consequently, the claims of Sections V to VII are stated for one or several independent per-directed-link faults on a stationary fabric, and we state the converse as plainly: under a correlated shift the ledger's absolute-floor test does not hold, and a relative passive test is the better tool. The uniform shift is the cleanest stress and the one the decision rule was predicted to fail; a partial correlated shift over a subset of links is untested and would sit between the first and second regimes. Section X discusses what a fix would need.

IX. COST

The passive detectors pay nothing on the wire, and the ledger does. This section reports the ledger's cost in the currency each baseline paper uses for its own, from the compiler's resource report on the switch's own software development kit, not from an estimate. Table VI presents it beside the numbers the two baseline papers report for themselves.

Wire. The witness is two bytes per packet, i.e., about 0.14 percent of a 1400-byte payload, computed as $2/1404$. The earlier four-byte version, which also carried a link identifier, cost 0.28 percent, the same order as the ± 0.25 percent flow-priority distortion SprayCheck reports for its own prioritized flows [7]. The stamp is stripped before host delivery, so the cost is bandwidth on every fabric hop and nothing at the endpoints.

State. Each directed sublink holds a 16-bit sequence register and a 32-bit observed-count register, six bytes, and the program tracks all 1024 sublinks continuously. SprayCheck's own state is under two kilobytes, but it instruments one flow at a time and takes about a minute to rotate to the next [7]; FlowPulse's paper does not state a memory figure [8]. The trade is continuous per-link coverage of the whole fabric for a fixed per-packet tag, against zero tag for a rotating sample.

Pipeline. Halving the wire cost was not free. Reconstructing the directed link at ingress from the port and the spray

field added two narrow tables, and the compiler placed them in a new stage. The program went from 11 to 12 ingress stages, which is the hardware ceiling of the Tofino 1, with 5 egress stages unchanged. It also went from 89 to 91 SRAM blocks and from 15 to 16 TCAM blocks. The compile was byte-for-byte identical on the laptop SDE 9.13.1 and the switch's 9.13.2, with zero errors and the same five warnings on both. The stage cost follows from a choice we made before compiling. To avoid destabilizing logic that other parts of the program rely on, we added narrow tables rather than widen the key of an existing multi-purpose port-role table. Consequently, the program now has no ingress stage headroom, and any future ingress addition must either remove a stage elsewhere or revisit that choice by folding the reconstruction key into the port-role table. We report this as the real cost of the two-byte witness rather than as a footnote, because a reviewer who weights pipeline stages over wire bytes can reasonably prefer the four-byte version.

X. DISCUSSION AND LIMITATIONS

Common-mode load. The failure of Section VIII is the ledger's largest limitation, and it is a property of the decision rule rather than of the witness. The registers still recover each link's loss exactly under a fleet-wide shift; what fails is the comparison against a floor that moves with the fault. A fix may need a guard that detects when the floor itself is drifting, or a corroboration gate that requires a link's ratio to stand out relative to its neighbours as well as relative to the floor, i.e., the relative normalization that lets SprayCheck hold in that regime. We leave that gate to future work, and we note that it is design work with its own evaluation, not a parameter change.

Restoration. The paper stops at naming a directed link. Systems that then quarantine the link and re-admit it, such as CorrOpt [5], Aegis [6], and REPS [3], re-admit on evidence from real or host-generated traffic, and a link that spraying has starved of traffic can supply no such evidence. The ledger's vantage exposes that gap but this paper does not close it; we cite the repair systems to bound what the measurement enables.

What the localization result is. The ledger's exact-localization score of 1.00 is close to definitional, and we say so where we report it. Given each directed link's own transmit count, a detector that flags a link has already localized it. The measured fact is that the passive arms, which lack that count, alias the uplink and downlink pair at low loss and cannot recover it, and that the aliasing is not a quirk of one baseline but a limit of a single receive-only vantage under spraying. The result should be read as the witness removing an information-structure limit, not as an inference advance.

The wire cost is a real trade. Two bytes per packet on every fabric hop is a cost the passive detectors never pay, and the halving from four bytes consumed the program's last ingress stage. An operator who weights pipeline headroom over wire bytes can prefer the four-byte version, and an operator who cannot tag packets at all cannot use the witness. We report both sides so that the trade can be made rather than assumed.

An open anomaly on unmeasured sublinks. During a 57-cycle soak of the two-byte program on the switch, the injected-loss recovery passed in every cycle, but two sublinks that were neither armed nor measured each showed one stamped packet with no matching arrival during clean traffic, a signature not seen across about 3 200 historical cycles of the four-byte program. Port-level media-access counters ruled out a physical-layer drop, and a fully idle window ruled out ambient noise; the cause is not known. Every number in this paper comes from armed, measured sublinks with baseline and post-traffic reads, and none is affected, but we record the anomaly rather than omit it.

Generality. The replay fabric has four leaves and eight spines with uniform spray, and the silicon fabric is a four-leaf, two-spine virtual topology on one switch. FlowPulse’s dilution grows with senders per port and SprayCheck’s intersection improves with leaves, so the crossover rates are properties of this fabric. The $1/p^2$ wall for a count-based test should not depend on the topology, since it follows from the spray noise alone. SprayCheck’s sensitivity was calibrated under its paper’s Poisson model; its real spraying has less noise, which would lower its curve by a constant factor. The budget of 160 million packets sets where the passive action rate collapses, not whether it does.

XI. RELATED WORK

Table VII positions the ledger against the systems closest to it on the axes this paper measures. We group them into four families.

In-fabric loss witnesses. NetSeer [9] inserts a per-neighbour sequence number at egress and treats an inconsecutive number at the downstream ingress as a drop, which is exactly the ledger’s encoding at four bytes; LinkGuardian [10] adds an era bit and a dummy packet and retransmits the lost packet across the link; LossRadar [11] stores packet digests in invertible Bloom filters at both ends of a link and decodes the difference, which its authors state works per packet only below about 0.1 percent loss; the Ultra Ethernet link-layer retry [2] places the sequence in the MAC; and RFC 6374 [15] pairs transmit and receive counters per measurement entity. These systems own the primitive this paper uses, and we claim none of it. Compared to this work, none of them reports what the witness buys or costs on a sprayed fabric at 10^{-3} to 10^{-4} , measured against the passive detectors built for that fabric.

Passive detectors for sprayed fabrics. SprayCheck [7] and FlowPulse [8] are the direct comparators of this paper and the state of the art for sprayed fabrics. Both are zero-byte, both are deployable on existing counters, and SprayCheck reports zero added network load. The ledger does not dominate them on added load, because it pays two bytes per packet and they pay nothing. What it buys, measured in Sections V and VI, is a detection cost that does not grow as $1/p$ and a localization that does not alias the uplink and downlink pair, and both follow from holding each directed link’s own transmit count. Under a common-mode shift, SprayCheck’s relative test can hold where the ledger’s absolute one cannot (Section VIII). OmniPath Ping [19] is concurrent work that addresses path

ambiguity under spraying with an in-network per-hop cache and active probes; it is a tracing tool rather than a cost study, and it owns the problem statement of active measurement under spraying.

Active probing and tomography. dShark [16] localizes drops, including silent ones, by analysing mirrored per-hop traces on hosts after the fact, at a cost of mirroring the traffic and of capture drops that can hide real drops; the ledger’s only edge over it is that two registers per sublink on a rerouteable resource are cheaper and act in place, not that it can find anything dShark cannot. NetBouncer [17], 007 [20], deTector [21], Pingmesh [22], and Everflow [23] probe or trace paths from hosts and infer per-link health; under per-packet spraying a probe cannot pin its path, so these systems localize to a path or a scalar per link rather than to a directed link. Bartolini et al. [18] give the identifiability bounds of Boolean tomography as a function of routing, which is the framing in which the aliasing limit of Section II-B sits; the ledger sidesteps the bound by adding information rather than by a better inference, and we make no identifiability claim.

Repair and restoration. CorrOpt [5] disables a corrupting link in both directions and re-enables it after repair; Aegis [6] quarantines, audits with host-generated traffic, and certifies; REPS [3] reroutes sprayed traffic away from a lossy path and re-explores it. All three re-admit on evidence from real or generated traffic, which a link that spraying has starved cannot supply. Compared to this work, they serve a different objective, the lifecycle after localization, and we cite them to bound what the measurement enables rather than to claim restoration. We have found no prior system that reports the head-to-head of Table VII’s last row against its SprayCheck and FlowPulse rows on a sprayed fabric, on silicon, at the low-loss regime, and that row is what this paper adds.

XII. CONCLUSION

We measured what a two-byte per-directed-link witness buys, and costs, on a packet-sprayed fabric against the two most recent passive detectors built for such fabrics, replayed from their own papers over one shared stream, and we measured the witness itself on an Intel Tofino switch. The witness detects a grayhole at a flat median of 22 million packets from 1.5 percent loss down to 10^{-4} with no false positive, where the passive cost grows to 114 million packets at 10^{-3} and exceeds a 160-million-packet budget at 10^{-4} . It names the exact directed link where the passive detectors return the uplink and downlink pair, it recovers injected loss exactly on silicon down to six packets in sixty thousand, and it costs about 0.14 percent of a 1400-byte payload plus one pipeline stage. The advantage widens with several independent faults and fails under a fleet-wide shift in loss, where a relative passive test is the better tool, so every claim is stated for independent faults on a stationary fabric. The witness is an existing primitive; the contribution is the measurement and its boundary. In future work, we intend to design and evaluate a corroboration gate that keeps the ledger’s exact localization under a common-mode shift.

TABLE VII

PRIOR SYSTEMS ON THE AXES THIS PAPER MEASURES. DL: DIRECTED LINK. “NOT ADDRESSED” MEANS THE PAPER DOES NOT TARGET THAT AXIS. THE LEDGER ROW RESTATES MEASUREMENTS FROM SECTIONS V TO IX; EVERY OTHER CELL IS TAKEN FROM THE CITED PAPER.

System	Primitive	Vantage	Per-DL	Under spraying	Low-loss detection cost	Localization under spraying	Restoration
NetSeer [9]	4 B egress per-DL sequence	in-fabric, passive	exact DL	not addressed	next packet, $\Theta(1)$; not measured under spray	exact DL, not spray-measured	no
LinkGuardian [10]	B sequence, era bit, dummy packet	in-fabric, passive	exact DL	not addressed	$\Theta(1)$	exact DL, not spray-measured	link-local retransmit
LossRadar [11]	invertible Bloom filters at both ends	in-fabric, passive digest	exact DL	not addressed	per-packet decode only below about 0.1% retry, $\Theta(1)$	exact DL below 0.1%; counts above	no
UEC LLR [2]	per-link sequence in the MAC	in-fabric, passive	exact DL	in MAC, not measured		exact DL, no spray number	link-layer retry
RFC 6374 [15]	TX/RX counter pairs	in-fabric, passive	if measured per DL	attribution fails	epoch-skew limited	none under spray	no
SprayCheck [7]	per-(leaf, spine) RX Z-test, cross-leaf intersection	in-fabric counts, passive	by intersection	yes	0.78 at 10^{-3} , 0.00 at 10^{-4} ; about $\Theta(1/p)$ (measured here)	exact DL at 1–1.5%; pair at $\leq 10^{-3}$ (measured here)	no
FlowPulse [8]	learned per-port load, 1% threshold	in-fabric + model, passive	per-port rule	uplink diluted by senders	fails at $\leq 10^{-3}$ (measured here)	port set or miss; misses uplinks (measured here)	no
dShark [16]	mirrored per-hop trace analysis	host, post hoc	hop via TTL	not a spray study	localizes blackholes; capture drops	hop from trace	no
NetBouncer [17]	active probes, scalar per link	host, active	per-link scalar	path pinning breaks	link is one resource	per link, no DL under spray	no
Bartolini et al. [18]	identifiability bound	theory	bound	owns the framing	bound, no cost curve	monitor placement	no
OmniPath Ping [19]	in-network per-hop cache, probes	in-fabric, active	per hop	yes (concurrent)	tracing, not a cost study	per hop	no
CorrOpt [5]	corruption rate, disable, repair	host + counters	per-link scalar	fixed routing	not a spray localizer	no	disable both directions, repair, re-enable
Aegis [6]	enhanced probe sizes	host, active	path	cannot reach spray-hidden links	64 B probes missed a >1 KB fault	path level	quarantine, audit, certify
REPS [3]	spray load balancer, reroute away	in-fabric balancer	not a localizer	yes, avoidance	no loss localization	no	reroute away, re-explore
This paper	2 B egress per-DL sequence, two registers	in-fabric, passive on stamped stream	exact DL	yes	1.00 at 10^{-3} and 10^{-4} , flat 22 M packets	exact DL, set size 1, both families	not claimed

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